Lab 1: Diodes and Rectifiers

Team Roles

Activity	**Student Name**
Circuit Construction	
Data Collection	
Data Analysis	
Answer to Question	
Lab Report Writing	

1. Objectives and Background

The objective of this lab was to **deepen the understanding of the current-voltage characteristics of diodes and the input-output characteristics of a rectifier**. Meanwhile, the **techniques of circuit construction** and data harvesting are exercised and demonstrated.

2. Equipment Used

Table 1: Equipment used in Lab 1

Equipment	Quantity
Bread board	1
1N4007-TP diode	1
100Ω Resister	1
10kΩ Resister	1
Power Supply	1
Digital Multimeter	1
Oscilloscope	1
Signal Generator	1
BNC to Alligator Clip Cable	1
BNC to BNC Clip Cable	1
Banana to Test Hook Test Lead	4

3. Experiment 1: Focusing on the diode

3.1 Brief Procedures

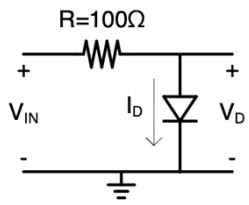

Figure 1 Circuit for experiment 1

The **circuit is constructed** as shown in Figure 1 and connected to the power and digital multimeter. The multimeter is linked to the circuit after the resistor is measured, when it is **apart from** the circuit. Then, **measure and record the current I_D and voltage V_D** while adjusting the input voltage V_{IN}. Finally, we calculate the demanded value, describe the necessary steps, and do error analysis.

3.2 Results and Analysis

Figure 2 shows the connection of the circuit on the breadboard by our group, where the resistor and diode are linked **in series**. When measuring the voltage, the multimeter should be connected to both sides of the diode, while it should be put into the circuit **in series** when observing the **current valu**

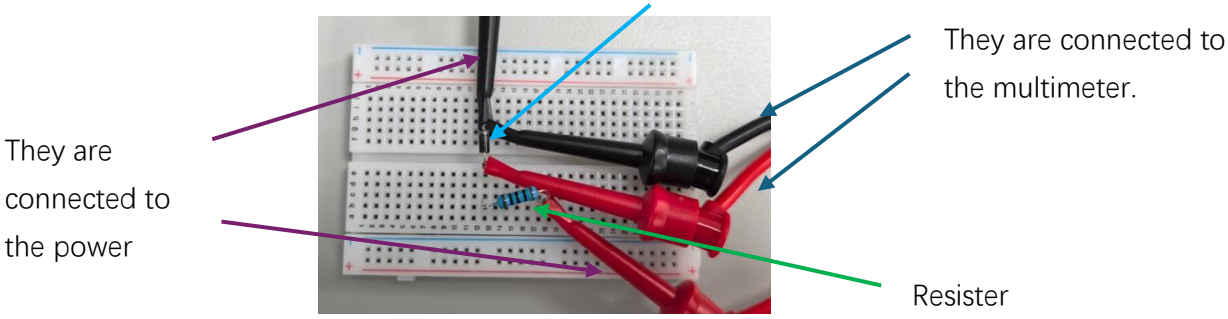

Figure 2 Circuit of experiment 1

The data from experiment 1 are demonstrated in Table 1. The cells with slashes indicate the meaninglessness of the data.

V_{IN} [V]	V_D [V]	I_{D1} [A]	I_{D2} [A]	Error
-5	-5	0		
0	-0.000764	0		
0.1	0.100018	0	4.15388E-09	
0.2	0.200851	1.60E-08	4.79544E-08	
0.3	0.30040	2.78E-07	5.3663E-07	
0.4	0.39930	4.69E-06	5.91131E-06	26.04%
0.5	0.49254	6.22E-05	5.67621E-05	8.74%
0.6	0.5613	0.000347	0.000300963	13.27%
0.7	0.60315	0.000922	0.000830698	9.90%
0.8	0.62879	0.001652	0.001547341	6.34%
0.9	0.64654	0.002471	0.002380124	3.68%
1	0.65976	0.003332	0.003280087	1.56%
2	0.71899	0.012658	0.013801667	9.04%
5	0.77105	0.041975	0.048801676	16.26%

Table 1: The current and voltage of Experiment 1

As illustrated in Table 1, the current and voltage of experiment 1, V_D and I_{D1}, are directly measured, while the data in the **last two columns** are obtained by calculation.

R (Ω)	99.377
I_S (A)	3.67E-10
n	1.65

Table 2 Calculated data

Using Excel.exe to analyze the measured data in Table 1, we get Table 2, which indicates the results of the **fitting process**. As learned in class, the equation for a diode can be represented as:

$$I_D = I_S\, e^{\frac{V_D}{nV_T}} \tag{1}$$

Then, using the parameters in Table 2, we calculate the last two columns in Table 1. Further, the error is computed by the formula:

$$\frac{|I_{D2} - I_{D1}|}{I_{D1}} \times 100\% \tag{2}$$

Most errors are below 10%, demonstrating the rationality of the experimental operation. Specifically, the error may come from the **measurement,** inevitably, due to the **accuracy of the devices**.

3.3 Answer to Lab Questions

3.3.1 A figure to explain your setup of Action #1.1

As shown in Figure 3, which is attached in Figure 2 Circuit of experiment 1, section 3.2. At this moment, the multimeter is used to measure the voltage on the diode.

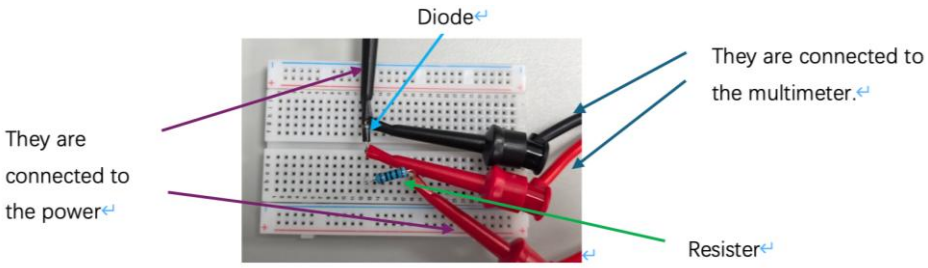

Figure 2 Circuit of experiment 1

Figure 3: Illustration of the setup in experiment 1

3.3.2 Discuss how you obtained your model (IS and n).

The formula (1) has shown us the characteristics of diodes. To simplify the calculation process, we take the logarithm of both sides of this formula and obtain:

$$\ln(I_D) = \ln(I_S) + \frac{V_D}{nV_T} \tag{3}$$

Then, the intercept can indicate the value of Is, and the slope demonstrates the

value of n with these pairs of data. Noted that V_T is the thermal voltage, which can be computed by the equation:

$$V_T = \frac{kT}{q} \tag{4}$$

Input the data in an Excel sheet, then we choose to fit the data using the function y=kx+b. Allowing the **parameters shown in the graph**, the fitting graph of formula (3) is a straight line, as shown in Figure 4.

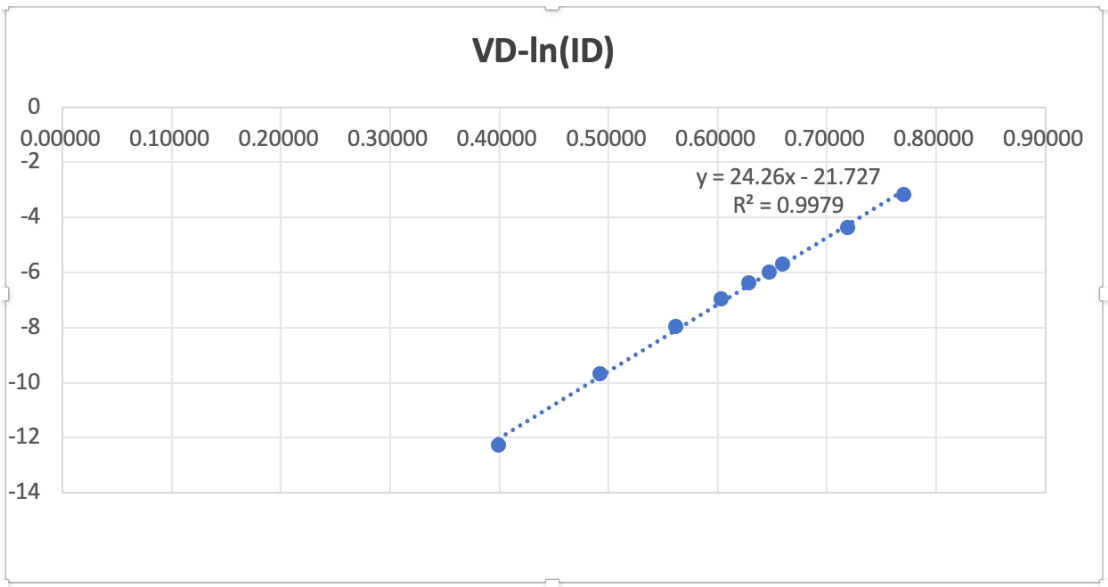

Figure 4: Fitting result of formula (3)

4. Experiment 2: Focus on the resistor

4.1 Brief Procedures

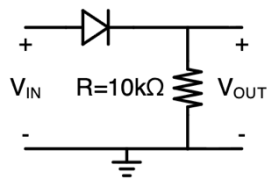

Figure 5 Circuit for experiment 2

Using the **constant-voltage-drop model** to analyze the target circuit as Figure 5 first. Then, constructed the circuit and connected it to the signal generator as well as the oscilloscope. Finally, analyze the circuit with an oscilloscope and record the related waveform.

4.2 Result and Analysis

Figure 6 shows a real connection on the breadboard. Similarly, when measuring the voltage, the multimeter should be connected to both sides of the diode, while in serial for the current value. Noted that the illustrated one is shown in section 4.3.

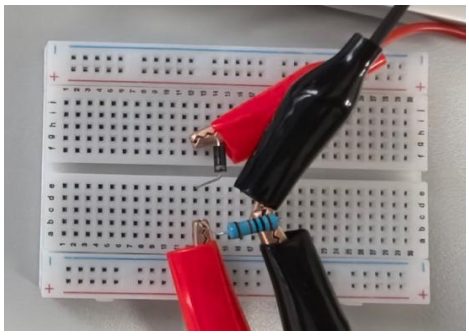

Figure 6 Circuit connection to experiment 2

The input is provided by the signal generator, which is set to offer a **sinusoidal signal** with a V_{pp} of ten and a frequency of 1 kHz. While giving the signal to the circuit, the same signal was provided to the oscilloscope simultaneously.

Hence, the waveform of V_{IN} and V_{OUT} in Figure 5 is shown in an oscilloscope, as shown in Figure 7. While the **V_{IN} is smaller than V_D**, the diode is off, which can be seen as an open circuit. Therefore, it is 0 during that period. Further, due to the v**oltage division of diodes**, the voltage on the resistor will be a bit lower than the input voltage.

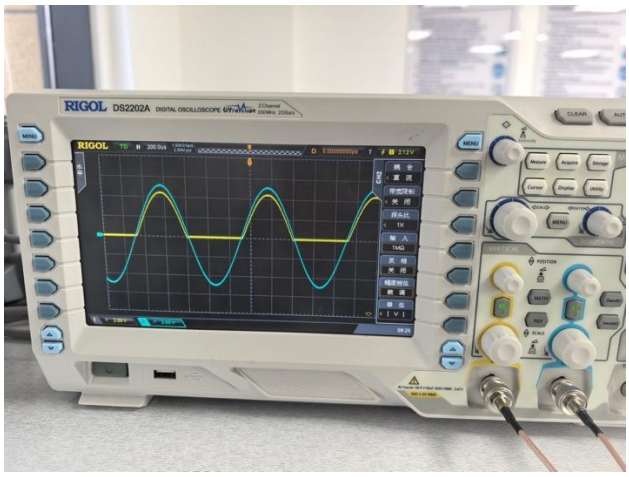

Figure 7: Input and output of the circuit

Then, change the oscilloscope to **X-Y model**, and the relationship between V_{IN} and V_{OUT} can be shown clearly, as illustrated in Figure 8.

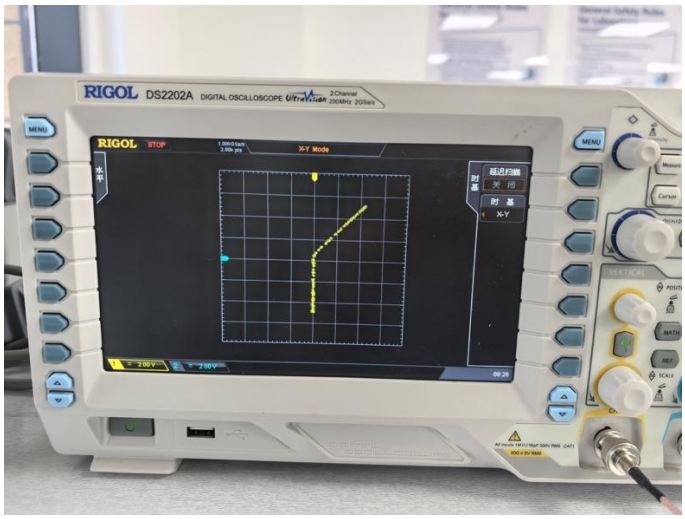

Figure 8 Relationship between V_{IN} and V_{OUT}

4.3 Answer to Lab Questions

4.3.1 Draw the transfer characteristics and output voltage in the time domain of Fig. 3.

Figure 9 is the theoretical curve of the transfer characteristics of the circuit.

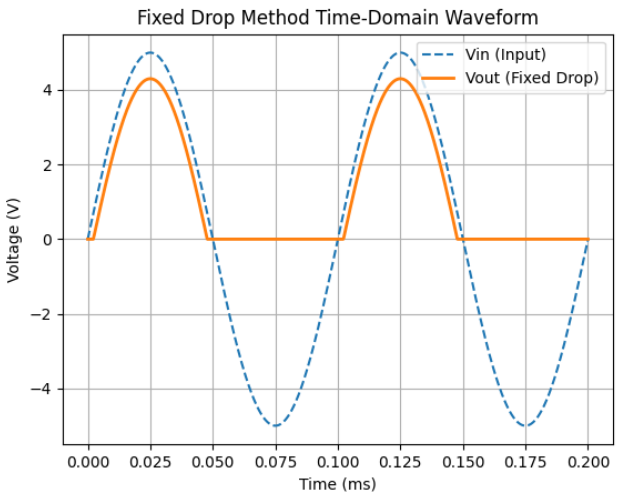

Figure 9 Transfer characteristics in the time domain

To draw the characteristics, we apply the constant-voltage-drop method, which approximates the diode to a **power source** about **0.7 volts** when it is conducting and open when it is not conducting. Then, we use the **relationship between V_{IN} and V_D** to determine whether it is conducting. Therefore, we got the equation:

$$V_{OUT} = \begin{cases} 10\sin(2\pi f t) - 10, & conducting \\ 0, & not\ conducting \end{cases} \quad (5)$$

Where f is 10 kHz. **Conducting means V_{IN} is bigger than V_{OUT}; otherwise, it is not conducting.**

4.3.2 A figure to explain your setup of Action #2.1 (Fig. 3).

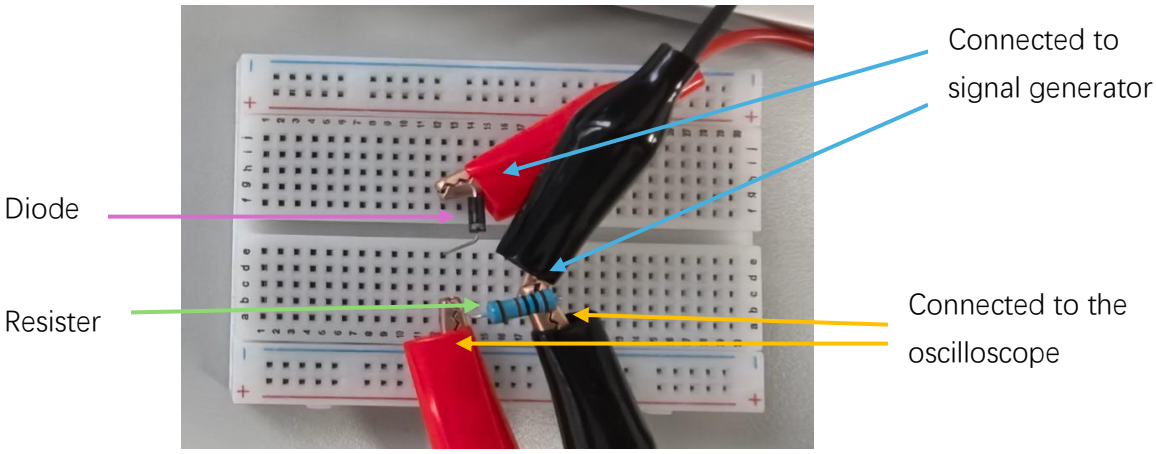

Figure 10: Illustrated setup figure.

As shown in Figure 10, the resistor and diode are connected in series, and the input is the signal from the signal generator. The oscilloscope is connected to both sides of the resistor, measuring the V_{OUT}.

4.3.3 A captured image of the oscilloscope that shows VIN(t) and V_{OUT}(t).

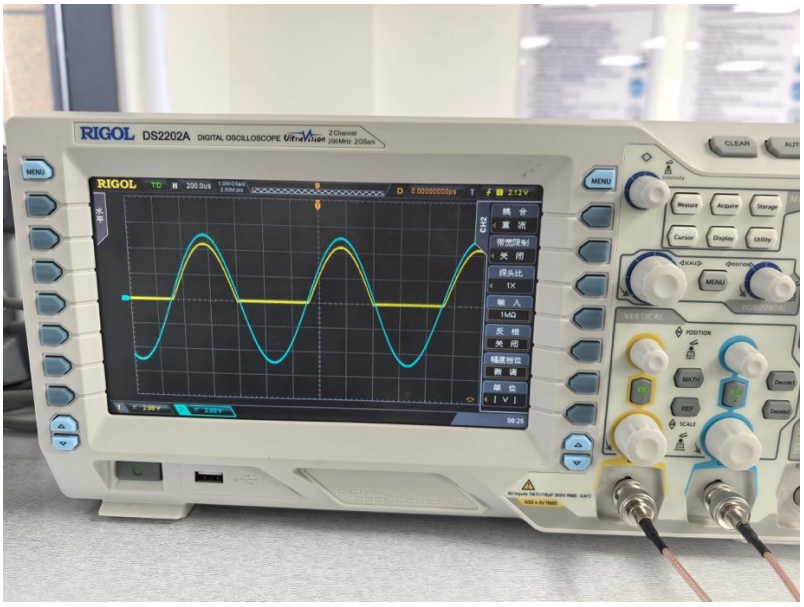

Figure 11 Oscilloscope of V_{IN} and V_{OUT}

4.4.4 Discuss the measurement result of Action #2.1.

Figure 11 shows the graph of V_{IN} and V_{OUT}, where the blue curve is V_{IN} and the yellow curve is V_{OUT}. As predicted in section 4.3.1, the V_{OUT} is a bit smaller than V_{IN} when it is **conducting**, and V_{OUT} becomes 0 when the **cutoff occurs**, due to the diode not conducting.

4.4.5 A captured image of the oscilloscope in the X-Y mode that displays the transfer characteristics of the circuit.

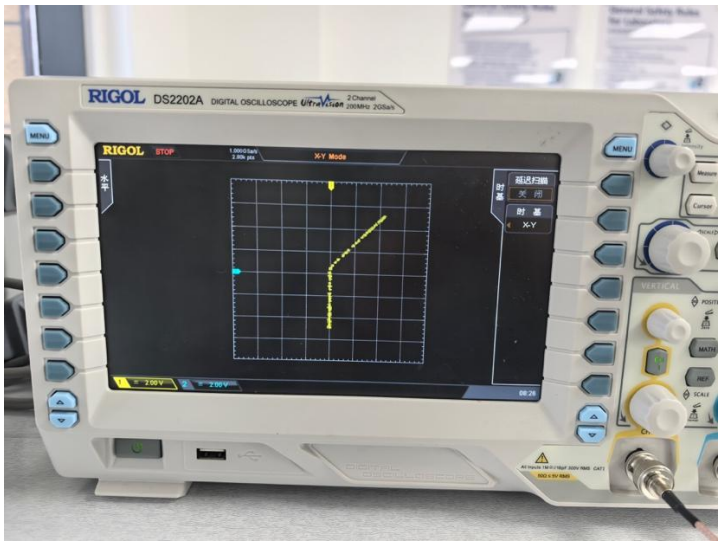

Figure 12 Image in the X-Y mode

4.4.6 Discuss the measurement result of Action #2.2.

Figure 12 is the relationship between V_{IN} and V_{OUT}, instead of the time field. The whole curve can be divided into **2 pieces**. The **vertical one** means **channel 1 (V_{OUT}) is nearly constant**, but **channel 2 (V_{IN}) is changing**, which indicates that V_{IN} is smaller than V_D and **the diode is cutoff**. So, **the V_{OUT} is 0**. For another piece, the two variables are both changing, indicating the diode is conducting, which forces the V_{OUT} to change with V_{IN}.

Based on the former mathematical deduction, the **corner** is V_{IN}, which approximates to 0.7 volts.

5. Conclusion

In this lab, we learned the operation to construct a circuit with a diode knowing

how to determine the direction of the diode. In addition, we exercised the knowledge to use the Excel software and became more familiar with the operation of fitting functions in Excel. Moreover, we learned the X-Y mode on the oscilloscope and knew more about the analysis process for a specific rectifier circuit, which helped us develop a deeper understanding of the diode.

6. References

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